

WIFI 蓝牙星闪工具移植



版本历史

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1. WIFI 测试工具移植

方式一： 在 WS73 SDK 里编译

(1) 选择所需的第三方组件进行下载，相关组件下载链接如下：

[libnl-3.5.0](#)、[libnl-3.5.0.tar.gz](#)、[wpa_supplicant-2.10](#) 和 [hostapd-2.10](#)

(2) 拷贝 ws73_sdk_linux_WS73_1.10.111/open_source/wpa_supplicant 下的补丁文件到 ws73_sdk_linux_WS73_1.10.111/open_source 下；将下载的压缩包拷贝到 ws73_sdk_linux_WS73_1.10.111/open_source 下；新建 wifi-tools-unpack.sh 文件。

```
mv open_source/wpa_supplicant/* open_source/  
vi open_source/wifi-tools-unpack.sh
```

```
#!/bin/bash

tar_packs=(hostapd-2.10.tar.gz libnl-3.5.0.tar.gz openssl-1.1.1n.tar.gz
wpa_supplicant-2.10.tar.gz)

src_dir=(hostapd-2.10 libnl-3.5.0 openssl-1.1.1n wpa_supplicant-2.10)

dst_dir=(hostapd libnl openssl wpa_supplicant)

length=${#tar_packs[@]}

for ((i=0; i<$length; i++))
do

    rm -rf ${dst_dir[$i]}

    echo "mkdir -p ${dst_dir[$i]}"

    mkdir -p ${dst_dir[$i]}

    echo "tar -xzf ${tar_packs[$i]} -C ${dst_dir[$i]} --strip-component=1 "

    tar -xf ${tar_packs[$i]} -C ${dst_dir[$i]} --strip-component=1

Done

cd wpa_supplicant

patch -p1 < ../wpa_supplicant_2_10_linux.patch
```

```
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source$ ls
hostapd-2.10.tar.gz  libnl-3.5.0.tar.gz  open_source.mk  wifi-tools-unpack.sh  wpa_supplicant-2.10.tar.gz
kernel              open_source_light.mk  openssl-1.1.1n.tar.gz  wpa_supplicant_2_10_linux.patch  wpa_supplicant_2_7_android_9.patch
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source$ cat wifi-tools-unpack.sh
#!/bin/bash

tar_packs=(hostapd-2.10.tar.gz libnl-3.5.0.tar.gz openssl-1.1.1n.tar.gz wpa_supplicant-2.10.tar.gz)
src_dir=(hostapd-2.10 libnl-3.5.0 openssl-1.1.1n wpa_supplicant-2.10)
dst_dir=(hostapd libnl openssl wpa_supplicant)
length=${#tar_packs[@]}

for ((i=0; i<$length; i++))
do
    rm -rf ${dst_dir[$i]}
    echo "mkdir -p ${dst_dir[$i]}"
    mkdir -p ${dst_dir[$i]}
    echo "tar -xzf ${tar_packs[$i]} -C ${dst_dir[$i]} --strip-component=1 "
    tar -xf ${tar_packs[$i]} -C ${dst_dir[$i]} --strip-component=1
done

cd wpa_supplicant
patch -p1 < ../wpa_supplicant_2_10_linux.patch
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source$
```

注意: wpa_supplicant_2_10_linux.patch 是基于 wpa_supplicant 2.10 生成的补丁文件, 若使用其他版本的开源组件, 可能会产生兼容性问题。

3.编译

```
chmod a+x wifi-tools-unpack.sh

./wifi-tools-unpack.sh

cd ..

make tools
```

ws73_default.config 配置文件添加 WSCFG_ARCH_NAME_TOOLS="aarch64"。

```
8 #
9 # Configure the compilation environment
10 #
11 WSCFG_USING_GCC=y
12 # WSCFG_USING_LLVM_CLANG is not set
13 WSCFG_CROSS_COMPILE="aarch64-mix210-linux-"
14 # WSCFG_KERNEL_DIR=""
15 WSCFG_EXTRA_CFLAGS=""
16 WSCFG_EXTRA_PARAMS=""
17 WSCFG_ARCH_ARM=y
18 # WSCFG_ARCH_CUSTOM is not set
19 WSCFG_ARCH_NAME="arm64"
20 WSCFG_ARCH_NAME_TOOLS="aarch64"
21 BOARD_ASIC=y
22 BOARD_ASIC_WIFI=y
23 WSCFG_BUS_USB=y
24 # WSCFG_BUS_USB is not set
25 # WSCFG_BUS_UART is not set
26 WSCFG_LINUX=y
```

修改 open_source/open_source.mk 文件。

```
77 libnl:
78 - cd $(LIBNL_DIR) && CC=$(WSCFG_CROSS_COMPILE)gcc ./configure --host=$(WSCFG_ARCH_NAME_TOOLS) linux --prefix=$(OUTPUT_BUILD_LIBNL_DIR)
79 - cd $(LIBNL_DIR) && make -j$(CPU_NUM) && make install -j$(CPU_NUM)
80 - @printf "${OK_COLOR}==== libnl build OK! =====${NO_COLOR}\n"
81
```

```
121 iperf:
122 - @cd $(IPERF_DIR) &&\
123 - CC=$(WSCFG_CROSS_COMPILE)gcc ./configure --host=$(WSCFG_ARCH_NAME_TOOLS) linux CFLAGS=-static\
124 - --enable-static LDFLAGS=-static --disable-shared --with-openssl=no
125 - cd $(IPERF_DIR) && make -j$(CPU_NUM)
126 - @printf "${OK_COLOR}==== iperf build OK! =====${NO_COLOR}\n"
127
```

解压压缩包。


```
sunshine@sunshine-vm: ~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source$ ./wifi-tools-unpack.sh
mkdir -p hostapd
tar -xzf hostapd-2.10.tar.gz -C hostapd --strip-component=1
mkdir -p libnl
tar -xzf libnl-3.5.0.tar.gz -C libnl --strip-component=1
mkdir -p openssl
tar -xzf openssl-1.1.1n.tar.gz -C openssl --strip-component=1
mkdir -p wpa_supplicant
tar -xzf wpa_supplicant-2.10.tar.gz -C wpa_supplicant --strip-component=1
patching file src/common/defs.h
patching file src/common/ieee802_11_defs.h
patching file src/common/wpa_common.c
patching file src/common/wpa_common.h
patching file src/drivers/driver.h
patching file src/drivers/driver_nl80211_android.c
patching file src/drivers/driver_nl80211.c
patching file src/drivers/driver_nl80211.h
patching file src/drivers/drivers.mak
patching file src/drivers/nl80211_copy.h
patching file src/l2_packet/l2_packet.h
patching file src/p2p/p2p.c
patching file src/utils/os.h
patching file src/utils/wpabuf.h
patching file src/utils/wpa_debug.c
patching file wpa_supplicant/config.c
patching file wpa_supplicant/config_file.c
patching file wpa_supplicant/config.h
patching file wpa_supplicant/config_ssid.h
patching file wpa_supplicant/ctrl_iface.c
patching file wpa_supplicant/defconfig
patching file wpa_supplicant/events.c
patching file wpa_supplicant/Makefile
patching file wpa_supplicant/sme.c
patching file wpa_supplicant/wapi/hash.h
patching file wpa_supplicant/wapi/types.h
patching file wpa_supplicant/wapi/wai_asn1.h
patching file wpa_supplicant/wapi/wai_call_back.h
patching file wpa_supplicant/wapi/wai_cert.h
patching file wpa_supplicant/wapi/wai_ec.h
patching file wpa_supplicant/wapi/wai_lib.h
patching file wpa_supplicant/wapi/wai_rtx.h
patching file wpa_supplicant/wapi/wai_sm.h
patching file wpa_supplicant/wapi/wapi.c
patching file wpa_supplicant/wapi/wapi.h
patching file wpa_supplicant/wpa_supplicant.c
patching file wpa_supplicant/wpa_supplicant.h
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source$

make[2]: 离开目录"/home/sunshine/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/open_source/wpa_supplicant/wpa_supplicant"
===== wpa_supplicant build OK! =====
===== libnl install OK! =====
===== openssl install OK! =====
===== wpa_supplicant install OK! =====
===== hostapd install OK! =====
make[1]: 离开目录"/home/sunshine/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111"
open source tools built success!
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111$ ls output/bin/open_source/
hostapd      hostapd.conf      libnl-3.so.200.26.0  libnl-route-3.so.200  wpa_cli      wpa_supplicant.conf
hostapd_cli  libcrypto.so.1.1  libnl-genl-3.so.200.26.0  libssl.so.1.1        wpa_supplicant
sunshine@sunshine-vm:~/Workspace/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111$
```

方式二： 单独提供的工具包

```
tar -xvf wifi_tools.tar.gz

cd wifi_tools

make
```

```
make[2]: Leaving directory '/home/ebaina/Workspace/opensource/wifi_tools/iperf'
make[1]: Leaving directory '/home/ebaina/Workspace/opensource/wifi_tools/iperf'
===== iperf build OK! =====
===== install OK! =====
ebaina@Ubuntu22:~/Workspace/opensource/wifi_tools$
```

生成目标文件在 output/bin 目录

```
ebaina@Ubuntu22:~/Workspace/opensource/wifi_tools$ ls output/bin/
hostapd      hostapd_cli      hostapd.conf      iperf3      wpa_cli      wpa_supplicant      wpa_supplicant.conf
ebaina@Ubuntu22:~/Workspace/opensource/wifi_tools$
```


2. 蓝牙测试工具移植

1. 安装系统依赖工具

```
sudo apt-get install gettext  
sudo apt-get install libglib2.0-dev  
sudo apt install python3-docutils  
sudo apt-get install libtool  
sudo apt-get install autoconf  
sudo apt install pkg-config  
sudo apt install cmake
```

2. 编译

```
tar -xvf bluetooth_tools.tar.gz  
cd bluetooth_tools  
make
```

```
make[1]: 离开目录“/home/sunshine/opensource/SS928/bluetooth_tools/bluez”  
~/opensource/SS928/bluetooth_tools  
printf "\033[32;01m===== bluez build OK! =====\033[0m\n”  
===== bluez build OK! =====  
===== install OK! =====  
sunshine@sunshine-vm:~/opensource/SS928/bluetooth_tools$
```

生成目标文件在 output/bin 目录

```
sunshine@sunshine-vm:~/opensource/SS928/bluetooth_tools$ ls output/bin/  
bluetoothctl bluetoothd dbus-daemon libdbus-1.so.3 libexpat.so.1 libglib-2.0.so.0 libncurses.so.6 libpcre.so.1 libreadline.so.8  
sunshine@sunshine-vm:~/opensource/SS928/bluetooth_tools$
```

3. 星闪测试工具获取

星闪的客户端与服务端源码分别在 [ws73_sdk_linux_WS73_1.10.111/application/sample/sle/sle_uuid/](https://www.ebaina.com/opensource/SS928/bluetooth_tools/bluez) 对应目录下，请根据官方提供的编译说明或本章节提供的编译方式操作。

- 官方 sample 编译说明：

[ws73_sdk_linux_WS73_1.10.111/application/sample/sle/sle_uuid/sle_uuid_client/ReadMe.md](https://www.ebaina.com/opensource/SS928/bluetooth_tools/bluez)

● client 端:

(1) 进入 `ws73_sdk_linux_WS73_1.10.111/application/sample/sle/sle_uuid/sle_uuid_client/lib/` 目录, 拷贝社区提供的 `libsle_host.a` 静态库到当前目录下。

静态库文件路径: 【易百纳】Euler Pi 2.0\09. 进阶功能开发\01.常用工具移植\02.星闪_WIFI_蓝牙工具移植\第三方依赖

(2) 修改 `ws73_sdk_linux_WS73_1.10.111/application/sample/sle/sle_uuid/sle_uuid_client` 目录下 `Makefile` 中的'CROSS'宏为主控使用的工具链。

```
1 # CROSS 为交叉编译工具链
2 CROSS = aarch64-mix210-linux-
3 CC = @echo " GCC   $@"; $(CROSS)gcc
4 LD = @echo " LD    $@"; $(CROSS)ld
5 AR = @echo " AR     $@"; $(CROSS)ar
6 RM = @echo " RM     $@"; rm -f
7 STRIP = @echo "STRIP $@"; $(CROSS)strip
```

修改 `ws73_sdk_linux_WS73_1.10.111/driver/platform/osal/include/osal_types.h`。

```
31 typedef void*          osal_pvoid;
32 typedef osal_u8         osal_byte;
33 typedef osal_byte*      osal_pbyte;
34 // typedef unsigned long long      uint64_t;
35 typedef unsigned char    uint8_t;
36 typedef unsigned short   uint16_t;
37 typedef unsigned int      uint32_t;
38
39 typedef signed char       int8_t;
40 typedef short             int16_t;
```

在 `sle_uuid_client` 目录下执行以下命令进行编译即可生产 `sle_client_sample` 可执行文件。

```
make clean

make all
```

```
<):sle_client_sample; <):src//sle_client.o src//sle_uuid_client.o
GCC      sle_client_sample
STRIP    sle_client_sample
ebaina@Ubuntu22:~/Workspace/HiEulerPi/HiEuler_PT/Peripherals_Driver/ws73_sdk_linux
inc lib Makefile ReadMe.md sle_client_sample src
```

● server 端:

(1) 进入 `ws73_sdk_linux_WS73_1.10.111/sample/sle/sle_uuid/sle_uuid_server/lib/` 目录, 拷贝社区提供的 `libsle_host.a` 静态库到当前目录下。

静态库文件路径: 【易百纳】Euler Pi 2.0\09. 进阶功能开发\01.常用工具移植\02.星闪_WIFI_蓝牙工具移植\第三方依赖

(2) 修改 `ws73_sdk_linux_WS73_1.10.111/application/sample/sle/sle_uuid/sle_uuid_server` 目录下 Makefile 中的'CROSS'宏为主控使用的工具链和

`ws73_sdk_linux_WS73_1.10.111/application/sample/sle/sle_uuid/sle_uuid_server/src` 目录下的 `sle_uuid_server.c` 中引用 `time.h` 头文件的路径。

```
1 # CROSS 为交叉编译工具链
2 CROSS = aarch64-mix210-linux-
3 CC = @echo " GCC   $@"; $(CROSS)gcc
4 LD = @echo " LD    $@"; $(CROSS)ld
5 AR = @echo " AR     $@"; $(CROSS)ar
6 RM = @echo " RM     $@"; rm -f
7 STRIP = @echo "STRIP $@"; $(CROSS)strip
```

```
1 /**
2  * Copyright (c) @CompanyNameMagicTag 2023-2023. All rights reserved.
3  * Description: SLE UUID Server Source.
4  */
5 #include <time.h>
6 #include "securec.h"
7 #include "sle_common.h"
8 #include "sle_errcode.h"
9 #include "sle_connection_manager.h"
10 #include "sle_device_discovery.h"
11 #include "sle_uuid_server_adv.h"
12 #include "sle_uuid_server.h"
13
```

修改 `ws73_sdk_linux_WS73_1.10.111/driver/platform/osal/include/osal_types.h`。

```
31 typedef void*          osal_pvoid;
32 typedef osal_u8         osal_byte;
33 typedef osal_byte*      osal_pbyte;
34 // typedef unsigned long long      uint64_t;
35 typedef unsigned char     uint8_t;
36 typedef unsigned short    uint16_t;
37 typedef unsigned int      uint32_t;
38
39 typedef signed char       int8_t;
40 typedef short             int16_t;
```

在 sle_uuid_server 目录执行以下命令进行编译即可生产 sle_server_sample 可执行文件。

```
make clean
```

```
make all
```

```
<):sle_server_sample; <):src//sle_uuid_server_adv.o src//sle_uuid_server.o sle_app.o
GCC      sle_server_sample
STRIP    sle_server_sample
ebaina@Ubuntu22:~/Workspace/HiEulerPI/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/
inc lib Makefile ReadMe.md sle_app.c sle_app.o sle_server_sample src
ebaina@Ubuntu22:~/Workspace/HiEulerPI/HiEuler_PI_Peripherals_Driver/ws73_sdk_linux_WS73_1.10.111/
```




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